main.py | Output

```python
def to_8bit_binary(num):
    return format(num & 0xFF, '08b')


def signed_value(binary):
    value = int(binary, 2)
    if value >= 128:
        value -= 256
    return value


def subtract(a, b):
    # Represent operands in 8-bit 2's complement
    bin_a = to_8bit_binary(a)
    bin_b = to_8bit_binary(b)

    # 2's complement of B
    complement_b = to_8bit_binary(~b + 1)

    # Perform subtraction using A + 2's complement of B
    result_unsigned = int(bin_a, 2) + int(complement_b, 2)

    # Keep only the lower 8 bits
    result_binary = format(result_unsigned & 0xFF, '08b')
    result_decimal = signed_value(result_binary)
```

Visualize | Run

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```
    result_decimal = signed_value(result_binary)

    # Overflow detection for subtraction
    overflow = ((a < 0) != (b < 0)) and ((result_decimal < 0) != (a < 0))

    print("\n----------------------------")
    print("Expression:", a, "-", "(" + str(b) + ")")
    print("----------------------------")
    print("First operand (8-bit):  ", bin_a)
    print("Second operand (8-bit):  ", bin_b)
    print("2's complement of second:  ", complement_b)
    print("Binary addition result:  ", bin(result_unsigned)[2:])
    print("Final 8-bit binary result:  ", result_binary)
    print("Final decimal result:  ", result_decimal)
    print("Arithmetic overflow:  ", "Yes" if overflow else "No")


# (a) 45 - (+19)
```

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```
    subtraction
33      overflow = ((a < 0) != (b <
    0)) and ((result_decimal < 0) !=
    (a < 0))
34
35      print("\n----------------------
    ----------")
36      print("Expression:", a, "-", "
    (" + str(b) + ")")
37      print("----------------------
    --------")
38      print("First operand (8-bit):
         ", bin_a)
39      print("Second operand (8-bit):
         ", bin_b)
40      print("2's complement of
    second:     ", complement_b)
41      print("Binary addition result:
         ", bin(result_unsigned)[2:])
42      print("Final 8-bit binary
    result:     ", result_binary)
43      print("Final decimal result:
          ", result_decimal)
44      print("Arithmetic overflow:
          ", "Yes" if overflow else
    "No")
45
46
47  # (a) 45 - (+19)
48  subtract(45, 19)
49
50  # (b) (-35) - (-12)
51  subtract(-35, -12)
```

Visualize | Run

main.py Output

```
----------------------------------
Expression: 45 - (19)
----------------------------------
First operand (8-bit):         00101101
Second operand (8-bit):        00010011
2's complement of second:      11101101
Binary addition result:
100011010
Final 8-bit binary result:
00011010
Final decimal result:           26
Arithmetic overflow:            No

----------------------------------
Expression: -35 - (-12)
----------------------------------
First operand (8-bit):         11011101
Second operand (8-bit):        11110100
2's complement of second:      00001100
Binary addition result:        11101001
Final 8-bit binary result:
11101001
Final decimal result:           -23
Arithmetic overflow:            No
```

Visualize Run